



RAJALAKSHMI
ENGINEERING COLLEGE

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Department of ECE

EC19603 - Problem Solving using AI and ML
Techniques (Mini Project)

2023-24- Even

III Year ECE A section

Project Design Contest – 03.05.2024



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TITLE: "AI-Powered Typing Tutor: Improving Accuracy and Efficiency"

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OUTLINE

- Abstract
- Objective
- Literature Survey
- Summary of Literature
- Proposed System & Novelty
- Block Diagram
- Hardware/Software requirements
- References

Abstract

In today's fast-paced digital landscape, where effective communication and productivity are paramount, proficient typing skills have become indispensable. However, despite the widespread reliance on keyboards for various tasks, many individuals encounter challenges in maintaining typing accuracy, often struggling with misspelled letters. Recognizing the significance of this issue, our project introduces a novel solution: AI-Powered Typing Tutor, underpinned by advanced deep learning methodologies.

The primary objective of our project is to develop an intelligent system capable of analyzing users' typing behaviors to discern recurrent patterns of inaccuracies, particularly focusing on frequently misspelled letters. By harnessing the power of Long Short-Term Memory (LSTM) neural networks, the Predictive Typing Assistant can efficiently process vast amounts of typing data, enabling it to identify subtle nuances and trends within users' input sequences. Leveraging this knowledge, the system then generates tailored exercises and future typing prompts strategically designed to address specific areas of weakness and enhance overall typing accuracy.

Through the utilization of cutting-edge deep learning techniques, our project endeavors to revolutionize typing proficiency by providing users with personalized and targeted support. By empowering individuals to overcome common typing challenges and refine their skills in a tailored manner, the Predictive Typing Assistant aims to foster a more efficient and productive digital experience for users across various domains and industries.

WELCOME TO THE FUTURE OF TYPING MASTERY !



Objective

The primary objective of this project is to develop a Predictive Typing Assistant that utilizes deep learning algorithms to improve typing accuracy by analyzing and addressing frequently misspelled letters. Through this system, users will receive personalized typing practice tailored to their individual weaknesses, enabling them to focus on areas where they most commonly make errors. By identifying patterns in users' typing behavior, the system aims to provide targeted feedback and suggestions, ultimately helping users to develop more precise and efficient typing skills.

Additionally, the project aims to enhance user engagement and satisfaction by offering a seamless and intuitive typing experience. By leveraging state-of-the-art deep learning techniques, the Predictive Typing Assistant will continuously adapt and improve based on user interaction, ensuring that the assistance provided remains relevant and effective over time. Ultimately, the goal is to empower users with the tools and insights needed to become proficient typists, thereby facilitating smoother and more productive digital communication experiences.

Literature Survey

TITLE	YEAR	AUTHOR	TECHNIQUE
My tutor is an AI: The effects of involvement and tutor type on perceived quality, perceived credibility, and use intention	2022	Chen, M., Liu, F., & Lee, Y.H.	This study investigates how AI-based tutors impact user perceptions of quality, credibility, and intention to use
Bidirectional typing	2021	Dunfield, J., & Krishnaswami, N.	This study discusses the development of bidirectional typing systems that enhance user interaction with text input devices.
Observations on typing from 136 million keystrokes	2018	Dhakai, V., Feit, A.M., Kristensson, P.O., & Oulasvirta, A.	This paper analyzes a large dataset of typing behaviors to understand common patterns and errors

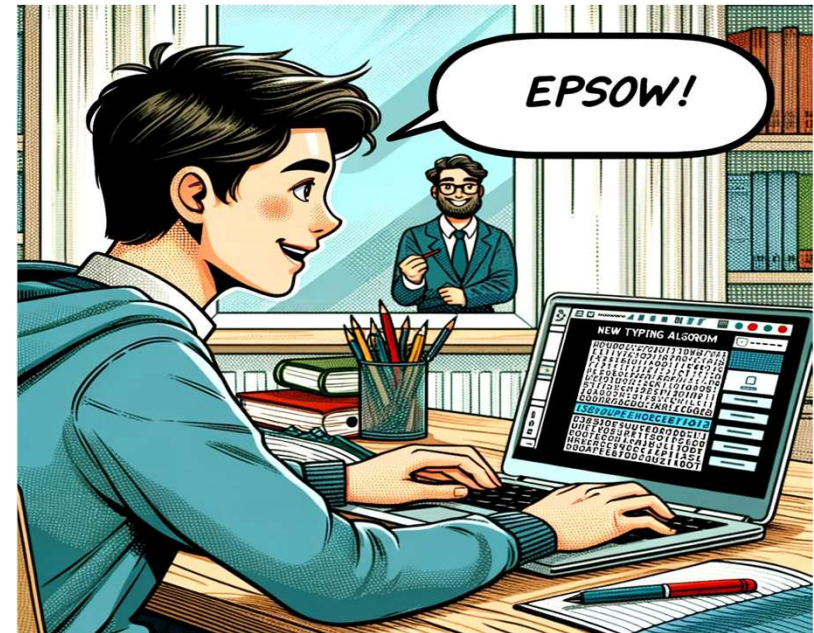
Summary of Literature

The importance of keyboarding skills in today's digital landscape, emphasizing the shift from typewriters to computer keyboards and the increasing demand for proficiency in typing. It underscores the significance of keyboarding education for both professional and non-professional workers, highlighting its implications for employability, earnings, and educational attainment. The text introduces the problem statement, research goals, and hypotheses, focusing on comparing the effectiveness of computer-based training versus conventional typewriter-based instruction in improving typing speed and accuracy.

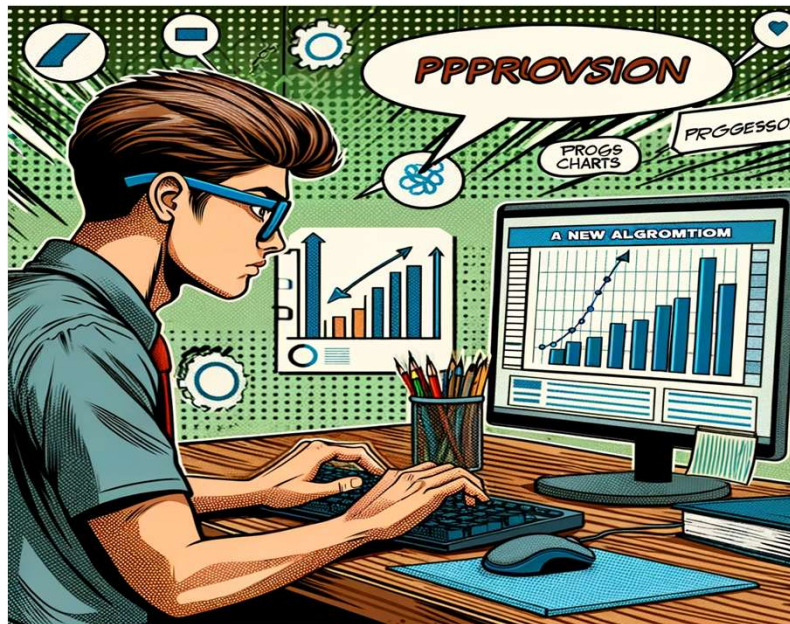
To integrate machine learning (ML) into this context, one could develop an advanced typing platform that leverages ML algorithms to personalize and optimize the learning experience. By analyzing user performance data, ML models can adapt the difficulty level of typing exercises in real-time, providing tailored recommendations and feedback to enhance skill development. Natural language processing (NLP) techniques can be employed to offer predictive text assistance and error analysis, helping users improve their typing speed and accuracy more efficiently. Additionally, ML algorithms can identify patterns in user behavior and performance to generate personalized training programs that address individual strengths and weaknesses. Overall, integrating ML capabilities into keyboarding education can enhance the effectiveness and efficiency of learning, empowering users to develop essential typing skills more rapidly and effectively in today's digital age.



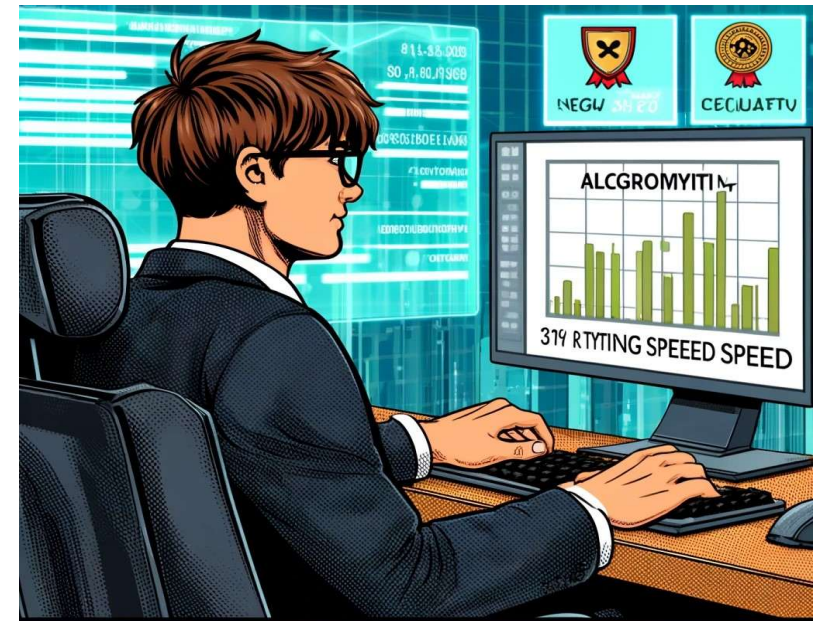
SCENARIO-1



SCENARIO-2



SCENARIO-3



SCENARIO-4

Proposed System

Data Collection and Analysis:

- The system begins by collecting user typing data, including both displayed sentences and corresponding user-typed responses. This data is then analyzed to identify patterns and trends in typing errors, with a focus on frequently misspelled letters.

Deep Learning Model Training:

- Leveraging Long Short-Term Memory (LSTM) neural networks, the system is trained on the collected data to learn and understand the underlying structures of typing sequences. The LSTM model is particularly well-suited for this task due to its ability to capture long-range dependencies and temporal patterns.

Pattern Recognition and Prediction:

- Once trained, the LSTM model is capable of recognizing common mistyped letters based on the input sequences. By analyzing contextual information and previous typing behavior, the system predicts the likelihood of mistyping specific characters in future typing sessions.

Generation of Tailored Typing Exercises:

- Using the insights gained from the LSTM model, the system generates customized typing exercises designed to target areas of weakness identified in the user's typing patterns. These exercises include sentences and prompts intentionally incorporating mistyped letters to facilitate focused practice and improvement.

User Feedback and Adaptation:

- As users engage with the typing exercises, the system continuously collects feedback on typing accuracy and performance. This feedback loop allows the system to adapt and refine its predictions over time, ensuring that the typing assistance provided remains relevant and effective for each user.

Evaluation and Monitoring:

- The system incorporates mechanisms for evaluating user progress and monitoring typing proficiency levels. Through comprehensive analytics and performance metrics, users can track their improvement and identify areas for further refinement.

Novelty in Proposed System

While existing typing assistance platforms may offer features to highlight misspelled letters, the proposed Predictive Typing Assistant introduces several novel capabilities that set it apart from traditional systems:

Dynamic Adjustment of Typing Challenges:

- Unlike static typing platforms that merely highlight misspelled letters, our system dynamically adjusts typing challenges based on users' performance and progress. By continuously analyzing typing patterns and adapting exercises in real-time, the Predictive Typing Assistant offers personalized and evolving support tailored to each user's needs.

Targeted Practice with Intentionally Incorporated Mistakes:

- While some platforms may provide general typing exercises, our system takes a more targeted approach by generating exercises that intentionally incorporate mistyped letters identified from users' typing history. This focused practice enables users to specifically address areas of weakness and accelerate their improvement.

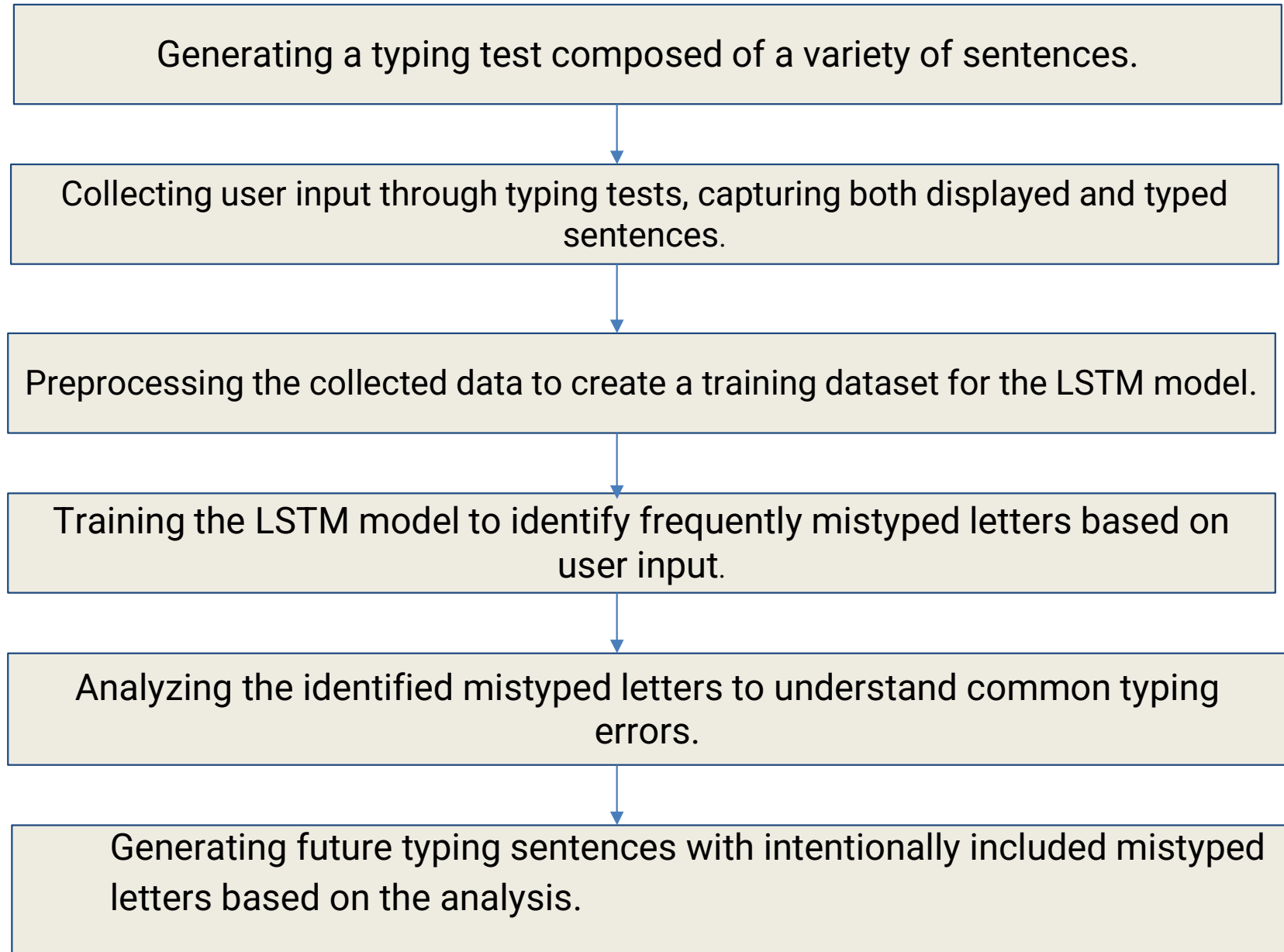
Continuous Learning and Adaptation:

- The Predictive Typing Assistant is designed to be a learning system that continuously evolves alongside the user. Through ongoing feedback and adaptation mechanisms, the system refines its predictions and recommendations, ensuring that typing assistance remains relevant and effective even as users' skills progress.

Comprehensive Performance Monitoring and Analytics:

- In addition to providing typing assistance, our system offers comprehensive performance monitoring and analytics features. Users can track their progress, identify trends in their typing accuracy, and gain insights into areas for further development, facilitating a deeper understanding of their typing habits and improvement trajectory.

Block Diagram



LIBRARIES USED

1. **NumPy (np)**: For numerical computing.
2. **Random**: For generating random numbers and shuffling sequences.
3. **Pandas (pd)**: For data manipulation and analysis.
4. **Matplotlib.pyplot (plt)**: For data visualization.
5. **Counter**: For counting occurrences of elements in a sequence.
6. **Sklearn.model_selection.train_test_split**: For splitting datasets into training and testing sets.
7. **TensorFlow.keras.models.Sequential**: For creating deep learning models.
8. **TensorFlow.keras.layers.LSTM**: Long Short-Term Memory (LSTM) layer for sequential data.
9. **TensorFlow.keras.layers.Dense**: Fully connected dense layer in neural networks.
10. **TensorFlow.keras.layers.Embedding**: For word embedding, converting words into dense vectors.
11. **TensorFlow.keras.preprocessing.text.Tokenizer**: For tokenizing text data.
12. **TensorFlow.keras.preprocessing.sequence.pad_sequences**: For padding sequences to ensure uniform length.

DATASETS USED

1. **Typing Accuracy Dataset (Comprehensive):**

1. This dataset contains information about words, including whether they were typed correctly or not. It's used to train the LSTM model to predict typing accuracy.
2. Columns: 'Word' (the word being typed), 'Correct' (1 if the word was typed correctly, 0 otherwise).

2. **Words and Sentences Dataset:**

1. This dataset contains lists of words and sentences used for typing tests.
2. Columns: 'Words' (list of words for typing tests), 'Sentences' (list of sentences for typing tests).

These datasets are loaded from CSV files and preprocessed before being used in the project.

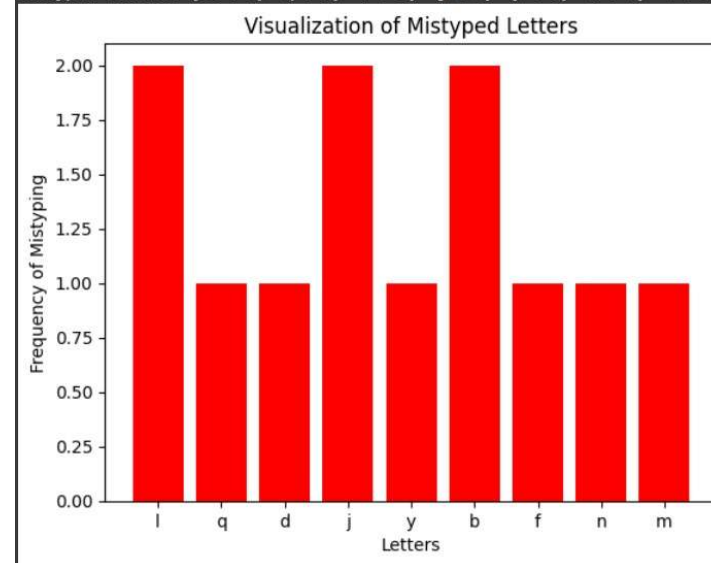
GLIMPSE OF THE PROJECT

```
Type the word: kel
kel
Type the word: lnr1
lnr1
Type the word: wmy
wmy
Type the word: jldfny
jldfny
Type the word: ltg

```

TEXT BOX FOR TYPING

```
Type the word: kel
kel
Type the word: lnr1
lnr1
Type the word: wmy
wmy
Type the word: jldfny
jldfny
Type the word: ltg
ltg
Type the word: qdmjyjf
eomyhif
Type the word: kbfbnmb
kobulw
Mistyped Letters: {'l': 2, 'q': 1, 'd': 1, 'j': 2, 'y': 1, 'b': 2, 'f': 1, 'n': 1, 'm': 1}
```



VISUALIZING THE INCORRECT LETTERS

References

1. Craighead, (2014, September). Typing: The key to computer literacy. Instructor. pp. 178 - 202.
1. Baars, A.I. and Swierstra, S.D., 2002, September. Typing dynamic typing. In *Proceedings of the seventh ACM SIGPLAN international conference on Functional programming* (pp. 157-166).
1. Tittelbach, D., Fields, L. and Alvero, A.M., 2008. Effects of performance feedback on typing speed and accuracy. *Journal of Organizational Behavior Management*, 27(4), pp.29-52.